CASE STUDY | PS-1 | FINANCE / ACCOUNTS PAYABLE

InvoiceDesk

From a vendor invoice PDF to a clear, reasoned payment decision, with every step visible.

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01 THE CUSTOMER

Who we are building for

Copperleaf Restaurant Group is a fictional mid-size company with 40 restaurant locations. It buys produce, packaging, cleaning services, kitchen-equipment rentals and IT support from a few dozen suppliers. Every supplier invoice arrives by email as a PDF, and a small Accounts Payable (AP) team decides which ones get paid.

~600	8-15 min	80-150 h	3
supplier invoices per month	of manual checking per invoice	of AP clerk time per month	costly mistakes that slip through on a tired Friday

02 THE PROBLEM

What happens today, and why it hurts

For every invoice, a clerk opens the PDF and works through the same checklist by hand:

- Find the matching **purchase order (PO)** in a spreadsheet, the record of what Copperleaf agreed to buy and at what price.
- Check the **goods receipt**, the record of what actually arrived, so we only pay for what we received.
- Confirm the supplier is **approved** and the payment is going to **their** bank account.
- Make sure the invoice **has not already been paid**.
- Decide: pay it, hold it and chase someone, or reject it.

The inputs are messy on both sides:

The invoice side	The procurement side
Every supplier formats invoices differently. Some PDFs have real text; others are phone scans.	Approved-vendor list, PO amounts and the tolerances finance will accept.
Line items may be itemised or bundled into one lump sum. Tax may be included in prices or listed separately.	One PO is often billed across several invoices, so each invoice must be checked against what was already billed.
The PO number may be printed, or only implied. The invoice number, date or total is sometimes missing.	Amounts are close but not exact, and someone has to decide whether 'close' is acceptable.

The expensive mistakes are not the obvious ones:

Mistake	How it happens	Why a person misses it
Paying twice	The same invoice is resent with a new number, or a scanned copy arrives after the digital one was paid.	The number is different, so a lookup by number finds nothing.
Overbilling across invoices	Each invoice against a PO looks fine, but together they exceed the PO.	Each invoice is checked alone, with no running total in sight.
Paying a fraudster	An invoice that is correct in every way, except the bank account has been swapped.	Nothing looks wrong, so nobody compares the bank details.

PROBLEM STATEMENT

Copperleaf's AP team spends 80-150 hours a month manually matching messy supplier invoices against purchase orders and goods receipts, and still risks duplicate payments, overbilling and bank-detail fraud. **We need a process that takes any invoice PDF, digital or scanned, one at a time or 50+ at once, and returns a clear decision of APPROVE, HOLD or REJECT, with every check behind it visible. For anything not approved, it should say who must act and what to tell them.**

What success looks like

- Clean invoices are approved with **no human touch**; people only see the exceptions.
- Every decision can be **explained** to an auditor in plain English, with the evidence behind it.
- The process **never silently approves** something it is unsure about.
- A month-end burst of **50+ invoices** is cleared in minutes, not days, without lowering accuracy.

03 THE PROPOSED SOLUTION

InvoiceDesk: an AP assistant that reads, checks and decides

InvoiceDesk takes an invoice PDF and runs it through the same checks an experienced AP clerk would, in about the time it takes to open the email. Each check is shown live as it runs, and the result is one of three decisions:

APPROVE	Every check passed with high confidence. Scheduled for payment, no human needed.
HOLD	Something needs a person. The invoice goes to the right owner (AP clerk, buyer, vendor or security) with the reason, the evidence and a drafted message.
REJECT	Certain it must not be paid: an exact duplicate of an invoice already processed, or a blocked supplier.

THE DESIGN PRINCIPLE: AI READS, CODE DECIDES

AI is excellent at reading messy documents and unreliable at arithmetic and judgment calls about money. So InvoiceDesk uses AI **only to read** the invoice. Every calculation, comparison and decision is made by plain, testable rules. **An AI model never approves a payment.** AI then writes the plain-English explanation of the result and drafts the follow-up email, but it cannot change the decision.

How it reads an invoice: two paths, because invoices come in two kinds

Invoice type	How it is read	Why
Digital PDF (has real text)	LiteParse reads the text locally, keeping the layout. Gemini turns that text into structured fields.	Faster and cheaper. It also enables a **grounding check**: every amount, the invoice number and the bank account must appear word for word in the PDF's own text, so a misread value cannot slip through.
Scanned PDF (image only)	Gemini reads the page image directly with its vision capability.	There is no text to read, so we cross-check with arithmetic instead: line items, tax and total must add up.

Every extracted field carries a confidence level and the exact text it came from. The AI's own confidence is never trusted alone: if the numbers add up, the reading is confirmed; if they do not, the invoice goes to a person.

The eight checks, shown live as they run

#	Check	What it does	If it fails
1	Intake	Fingerprints the file and detects whether each page is digital or scanned.	Unreadable file: reject
2	Read	Extracts the vendor, bank account, invoice number, dates, PO number, line items, tax and total.	Hold for AP clerk
3	Validate	Checks the arithmetic, works out the tax when prices already include it, and fills safe gaps (a missing total is calculated from the lines, with a warning).	Hold; ask vendor to correct
4	Vendor	Matches the supplier to the approved list (by tax ID or name, allowing for spelling variations) and compares the bank account with the one on file.	Unknown vendor: hold for onboarding. Blocked: reject. Bank changed: hold for security
5	PO match	Uses the printed PO number, or finds the right PO from the line items when none is printed.	Hold for buyer or AP clerk
6	Three-way match	Compares each line's price with the PO (within tolerance) and its quantity with what was received, and checks the PO's running balance.	Hold for buyer, with suggested fix
7	Duplicate check	Looks for an exact repeat (same vendor and invoice number) and a disguised one (same vendor, amount, items and dates).	Exact: reject. Disguised: hold
8	Decision	Combines all results, assigns an owner, and writes the explanation and the draft email.	n/a

All eight checks always run, even after a failure, so a reviewer sees every problem at once rather than fixing one and discovering the next.

04 SCALE

Handling 50+ invoices at once

Invoices arrive in bursts: an inbox dump on Monday morning or the rush at month-end close. A clerk works through them one at a time. InvoiceDesk accepts a whole batch and processes it in parallel, with one important safeguard.

Phase	How it runs	Why
Reading (checks 1-3)	**In parallel**, up to 8 invoices at once (adjustable to the AI provider's rate limit).	Reading takes seconds per invoice and each invoice is independent, so this is where parallelism saves time.

Phase	How it runs	Why
Deciding (checks 4-8)	**One at a time, in the order the invoices arrived.**	These checks share data: PO balances and the record of what has already been paid. Deciding takes milliseconds, so this costs nothing.

WHY THE DECIDING PHASE MUST NOT RUN IN PARALLEL

PO-1043 has $12,000 left. Two invoices of $9,000 and $6,000 arrive in the same batch. If both were decided at the same moment, each would see "$12,000 remaining" and both would be approved: a $3,000 overpayment caused by speed. In the same way, two copies of one invoice in a batch would both pass the duplicate check. Reading in parallel and deciding in order gives the speed without the risk. The results are also the same on every run, which auditors need.

Target: a 52-invoice month-end batch cleared in about 1-2 minutes, against roughly 10 hours by hand. The screen shows both numbers side by side as the batch runs.

05 EDGE CASES

Four scenarios that show the judgment behind the design

We chose these because each one defeats a simple rule, and because two of them end in **approval**. A system that holds everything is as useless as one that approves everything.

Scenario	Why it is hard	What InvoiceDesk does	Result
1. One PO split across invoices, with an overrun	PO-1043 is $50,000. Two earlier invoices billed $38,000. Invoice 3 is $15,000. Each invoice alone looks fine.	Keeps a running total per PO. Sees the PO would be exceeded by $3,000 (6%). Suggests paying the $12,000 remaining and requesting a $3,000 credit note.	HOLD
2. A duplicate in disguise	A scanned copy of an invoice that was already paid, resent under a new invoice number. The file and the number are both different.	Compares content, not numbers: same vendor, amount, line items and a nearby date. Shows the original side by side. Holds rather than rejects, because it cannot be certain.	HOLD
3. Bank details changed	The invoice is perfect: approved vendor, correct PO, goods received. Only the bank account differs from the one on file.	Flags it to security as possible payment fraud and instructs that the change be verified by calling the contact **on file**, never the one on the invoice.	HOLD
4. A messy scan that still gets approved	A skewed phone scan, no PO number printed, and prices that already include tax.	Finds the right PO from the line items, works out the tax, confirms totals are within tolerance, and approves, with notes explaining each inference.	APPROVE

Also covered (18 test scenarios in total)

Scenario	Expected result
Clean invoices from different suppliers and layouts	APPROVE
Price 1.5% above PO (within tolerance)	APPROVE
Invoice with no total printed (calculated from the lines)	APPROVE
IT services billed as one bundled line against a monthly retainer	APPROVE
Price 6% above PO	HOLD
Billed for 100 units when only 80 were received	HOLD
No invoice number	HOLD
Line items do not add up to the total	HOLD
References a closed PO	HOLD
Supplier not on the approved list (onboarding request drafted)	HOLD
Exact duplicate of an invoice already processed	REJECT
Supplier blocked by procurement	REJECT

Live demo bonus: the interviewer can upload any invoice of their own. It will come from an unknown supplier, so it is held with a drafted onboarding request: an unexpected input handled gracefully.

06 WHAT THE AP TEAM SEES

The product

Screen	What it shows
New run	Drag and drop one PDF or fifty, or pick a prepared demo invoice.
Live run	The invoice on the left. The eight checks on the right, each turning green, amber or red as it runs, with the evidence behind it. A decision card at the bottom with the reason, owner and drafted email.
Batch view	A tile per invoice filling in live, running counts of approved, held and rejected, and elapsed time compared with the manual estimate.
Dashboard	Invoices processed, the share approved with no human touch, average processing time, and **money stopped** (duplicates, overruns and suspected fraud), plus the full run history.
Procurement data	Vendors, POs with billed-to-date balances, goods receipts, and the editable tolerance policy. Change a tolerance and rerun to see the rules respond.
Test suite and assumptions	One click runs every scenario and shows expected vs actual results with a pass rate. The assumptions page lists every judgment call we made.

07 PROOF

How we show it works, not just say it

- **18 scenario invoices** in four different layouts, a third of them scanned, each with a known correct answer. Each runs on fresh data.
- **A 52-invoice month-end batch** containing deliberate traps: a duplicate pair, two invoices that together overrun a PO, a bank change, a blocked and an unknown supplier.

- **Rule tests** for every check, run without the AI, so the rules themselves are proven independently.
- **A concurrency test** runs the PO-overrun trap 20 times with random timing and confirms the second invoice is held every time.

08 IMPACT

What the business gets

Measure	Today	With InvoiceDesk
Time per clean invoice	8-15 minutes	Seconds, with no human touch
Month-end batch of 50	A day or more	About 1-2 minutes
Duplicate payments	Caught only if the number matches	Caught by content, even with a new number or as a rescan
Overbilling across invoices	Rarely caught	Running total per PO, checked on every invoice
Bank-detail fraud	Depends on someone noticing	Every invoice compared with the vendor's account on file
Audit trail	Whatever is in someone's inbox	Every check, its evidence and its decision, stored and replayable

09 ASSUMPTIONS

Judgment calls made for this case study

- Amounts are in USD, and an invoice references at most one PO.
- Tolerances: price within 2% per line; total within 1% or $50, whichever is greater. All are editable.
- The vendor list is the source of truth for bank details; changes are verified by phone using the contact on file.
- A probable duplicate is held, not rejected. Only an exact repeat (same vendor and invoice number) is rejected automatically.
- Services (IT, cleaning) are treated as received. Held invoices do not use up a PO's balance.
- In a batch, invoices are decided in the order they arrived. A clerk takes about 12 minutes per invoice (for the time-saved figure).

10 NEXT STEPS

Deliberately out of scope for this week

- Pulling invoices straight from the AP mailbox, instead of uploading them.
- Highlighting on the PDF exactly where each value was read.
- Partial-approval actions (today the suggested fix is shown as text), multiple currencies and ERP posting.
- Sign-in and role-based access for AP, buyers and security.